

Junior Data Engineer - Challenge

Hello, and congratulations on reaching this stage of our process! This phase is a practical challenge, and it's meant to reflect the kind of work you'd actually do here: pulling data from an API, cleaning it up, working with exchange rates, and automating a small pipeline end to end. It's time to dig in and do some real ETL!

Thanks for taking the time to work through this - we know it's a real investment on your part, and we appreciate it. This is meant to feel like a small real project, not a quiz, so make it yours: use whatever tools and approach you'd naturally reach for.

Specs

Build a small end-to-end pipeline around the objectives below. This is intentionally open-ended - how you structure it and how you handle data issues is part of what we're evaluating. This project should exhibit both your Python and SQL skills.

1. **Ingest.** Pull the *orders_raw* table from this endpoint:

https://jzozteoirwfczcccltcdrr.supabase.co/rest/v1/orders_raw?apikey=sb_publishable_Xwjiw--qkKcbMuSbKd6l2w_wN9mpNTv

Store it in a SQL database of your choice (Supabase has a generous free tier if you don't already have one).

2. **Clean.** Create a new table *orders_clean* based on *orders_raw*, cleaned of any data inconsistencies you find. Use your best judgment on handling any data issues you find.
3. **Exchange rates.** Pull daily FX rates from a free source such as frankfurter.dev or exchangerate.host and store them in your database.
4. **Customer spend in EUR.** Create a table starting from *orders_clean* that shows total amount spent by each customer in EUR. Where the order currency isn't EUR, convert using the most appropriate FX rate for the date in *fx_reference_date*.
5. **Country/category breakdown.** Create a table showing total EUR revenue by country for orders where category is "Books" or "Electronics", but only include countries where that combined revenue exceeds €40,000, ranked by revenue.
6. **Automate.** Set the tables from steps 4 and 5 to refresh daily. For this exercise, the *orders_raw* data is static (unlike real life, where new orders come in daily), but some *fx_reference_date* values are set in the future in order to simulate some change. It's fine to pause or tear down this automation after 3-5 days - we don't need it running indefinitely, and you shouldn't incur any real cost keeping it alive.
7. **Writeup.** A short summary covering:
 - What data issues you found in *orders_raw*, how you handled them, and why you chose those solutions
 - How you'd monitor this in production - if the daily job silently failed, how would you find out?
 - Your AI usage: which tools, what you kept vs. changed
8. **Submit.** Add aqurate-careers as a collaborator on your GitHub repo, then submit [this short form](#) with a link to your repo. For any other communication or access issues, contact careers@aqurate.ai.

Do not get discouraged in case you find some points difficult - try to solve the rest, if possible. Our aim here is to understand how you think about solving the problems, and in case you don't succeed, understanding how you have tried to overcome the hurdles is equally important!

Submit your application before **01/09/2026 (23:59 RO)** by providing access to the GitHub repo and submitting the form above - no other confirmation is needed.

We'll review applications by 08/09/2026 and inform you of the result and next steps via email.

Good luck - we're looking forward to seeing what you build!